

Measure-Weighted Sampling: Grand Experiment Update

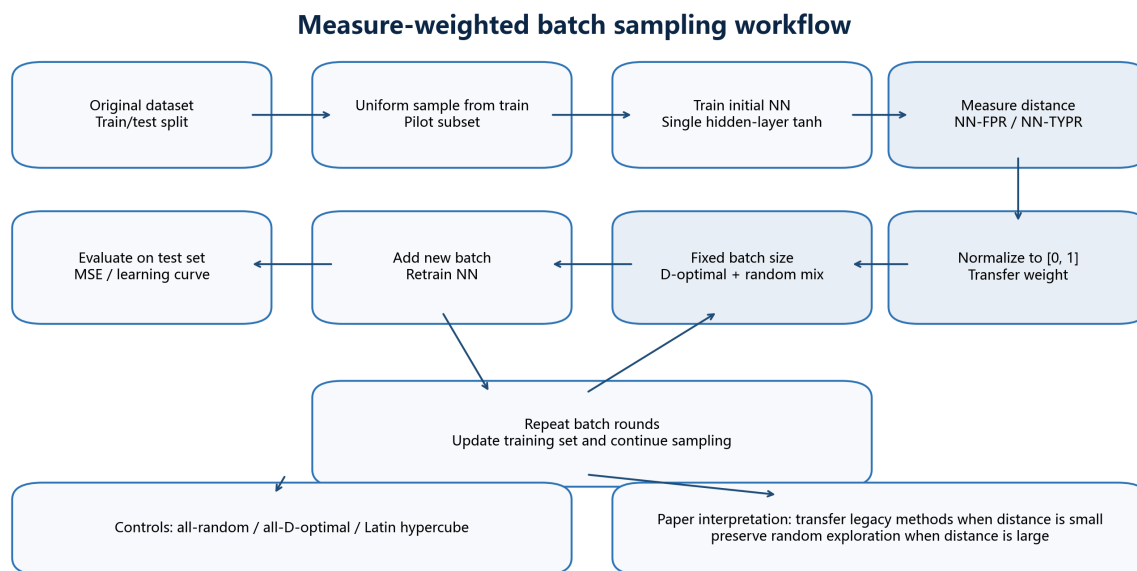
Using NN-PR/Taylor-PR distance as an operational gate for D-optimal transfer

Scale	n=10000; p=20 and p=50; five dataset families; five data seeds; two pilot seeds
Strategies	Measure-weighted, random, D-optimal, and Latin hypercube
Batch design	5% uniform pilot, 500 new points per round, six batch updates
Main rule	$\text{mean}(d_{\text{FPR}}, d_{\text{TYPR}})$ mapped to a 30%-95% D-optimal fraction

Main finding The distance measure is best interpreted as a transfer gate. Small NN-PR/Taylor-PR distance justifies more D-optimal transfer; large distance preserves random exploration and reduces negative transfer risk.

1. Workflow

In real applications the distance between a trained NN and a PR model is not known in advance. The workflow therefore starts with a uniform pilot sample, trains an initial NN, measures NN-FPR and NN-Taylor-PR distances, normalizes the distance, and uses the resulting weight to choose the D-optimal/random mix in the next batch.



Larger weight means more D-optimal points in the next batch; smaller weight preserves more random exploration.

Figure 1. Measure-weighted batch sampling workflow

2. Experimental Design

Each dataset is split into 75% training and 25% testing; 5% of the training split is used as a uniform pilot subset.

A single-hidden-layer tanh NN is trained on the pilot subset; NN-FPR and NN-Taylor-PR distances are measured on pilot validation points.

The main rule uses $\text{mean}(d_{\text{FPR}}, d_{\text{TYPR}})$, not $\text{min}(\dots)$, so high-dimensional false positives do not trigger excessive D-optimal transfer.

Each batch adds 500 points. The measure-weighted policy mixes D-optimal and random points; controls use all-random, all-D-optimal, and Latin hypercube selection.

3. Main Results

p	Strategy	Final MSE	Gain vs random	Positive rate	Runs
20	Latin hypercube	0.004811	5.79%	76.0%	50
20	D-optimal	0.004986	4.11%	76.0%	50
20	Measure-weighted	0.004993	4.40%	72.0%	50
20	Random	0.005113	0.00%	0.0%	50
50	Latin hypercube	0.018574	0.17%	46.0%	50
50	Random	0.018808	0.00%	0.0%	50
50	Measure-weighted	0.019143	0.50%	54.0%	50
50	D-optimal	0.019594	-1.85%	48.0%	50

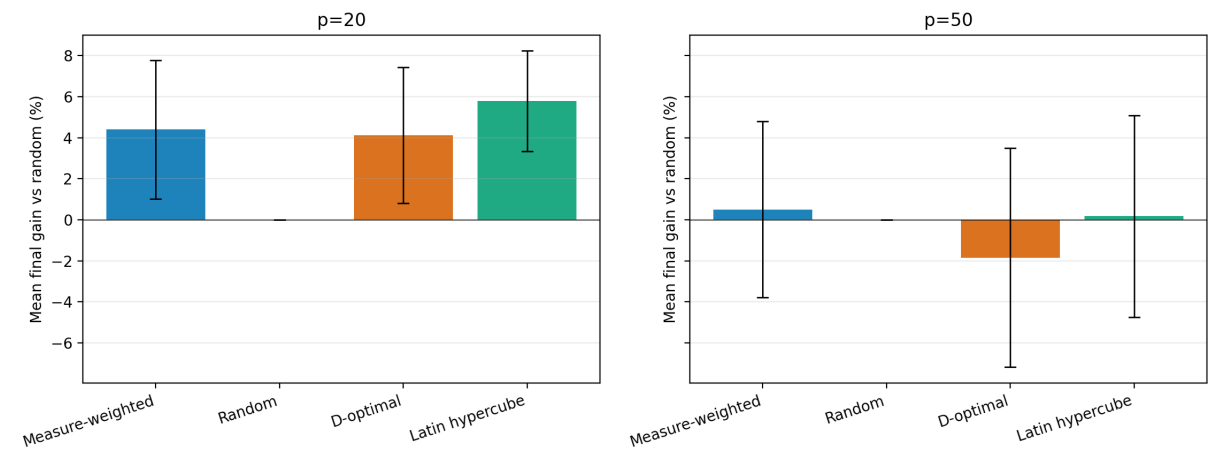


Figure 2. Final gain versus random sampling by dimension

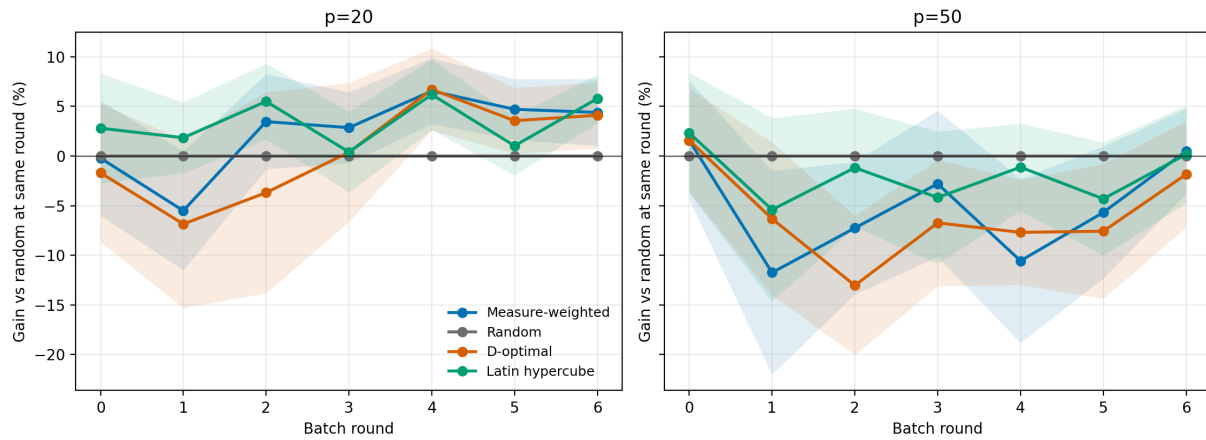


Figure 3. Learning-curve gain over multiple batch rounds

4. Case-Level Results

The measure-weighted policy is positive on most $p=20$ cases and mitigates the $p=50$ failure of all-D-optimal sampling. It is strongest on local-interior structure and remains useful in high-frequency stress tests, although Latin hypercube can still be stronger when global coverage dominates.

p	Dataset	Measure-weighted MSE	Gain vs random	Positive rate
20	High-frequency	0.010106	-9.52%	40.0%
20	Local interior	0.003893	5.82%	80.0%
20	Quadratic	0.007036	8.24%	90.0%
20	Smooth	0.002244	8.96%	90.0%
20	Strong nonlinear	0.001685	8.48%	60.0%
50	High-frequency	0.022579	0.24%	60.0%
50	Local interior	0.008042	13.55%	80.0%
50	Quadratic	0.049157	-8.05%	30.0%
50	Smooth	0.010861	-2.91%	30.0%
50	Strong nonlinear	0.005076	-0.33%	70.0%

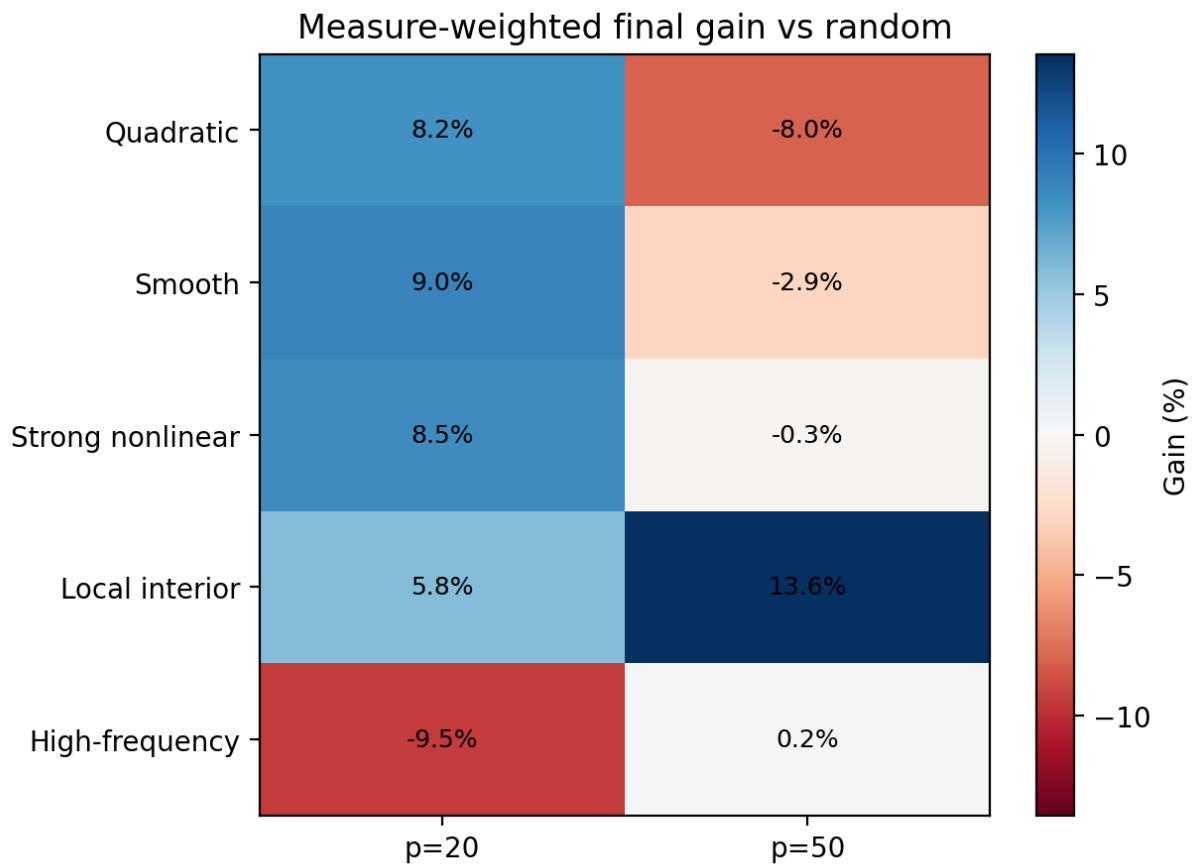


Figure 4. Case-level gain of the measure-weighted policy

5. Weight Calibration

The learned weights are not hard-coded by dataset label. They change with the pilot run. This is the operational value of the measure: it converts an observed NN-PR/Taylor-PR relation into a concrete sampling fraction.

p	Dataset	Mean FPR distance	Mean TYPR distance	Mean D-opt fraction
20	High-frequency	0.22958	0.94323	0.323
20	Local interior	0.08290	0.90947	0.551
20	Quadratic	0.04888	0.92314	0.579
20	Smooth	0.01406	0.78233	0.779
20	Strong nonlinear	0.01903	0.80521	0.749
50	High-frequency	0.01115	0.87244	0.377
50	Local interior	0.00287	0.69602	0.545
50	Quadratic	0.00381	0.67319	0.578

p	Dataset	Mean FPR distance	Mean TYPR distance	Mean D-opt fraction
50	Smooth	0.00507	0.53331	0.694
50	Strong nonlinear	0.00797	0.42922	0.815

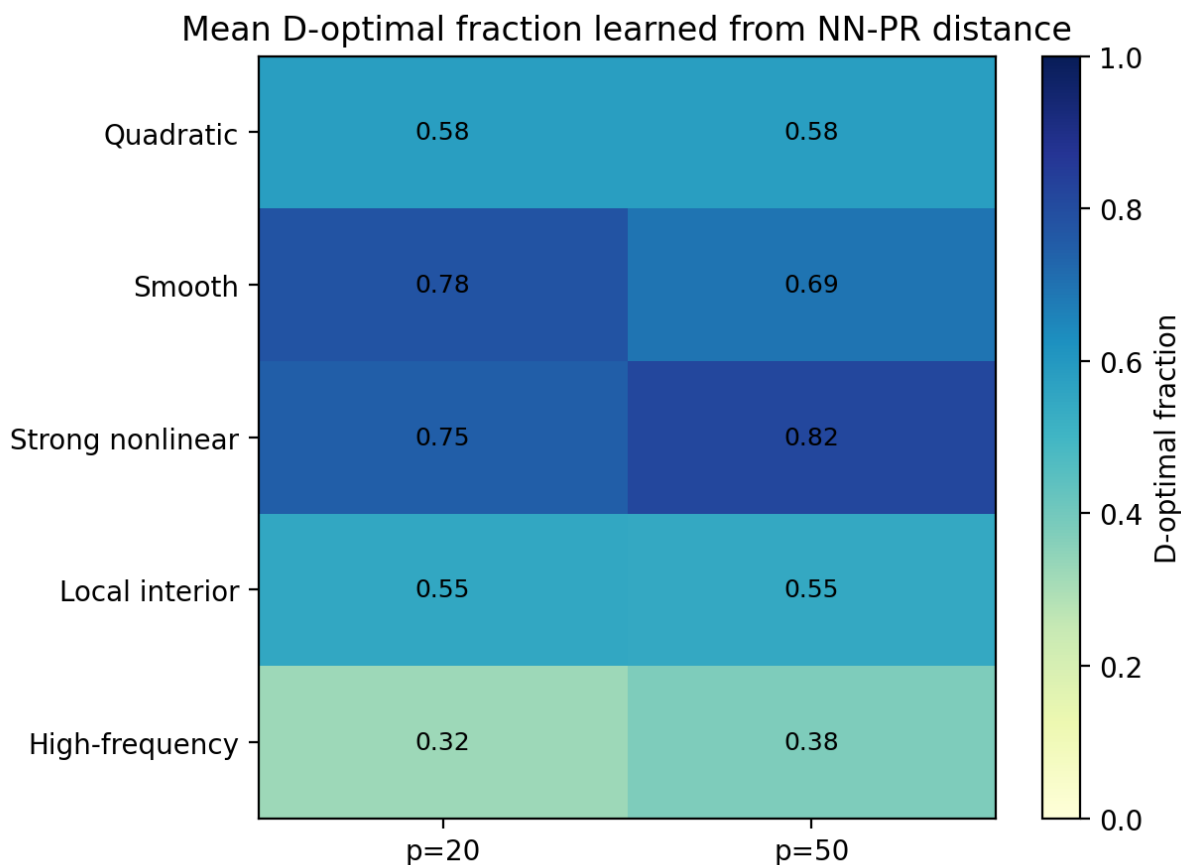


Figure 5. Mean D-optimal fraction learned from the distance measure

6. Sensitivity and Boundary Conditions

Important correction The earlier $\min(d_{\text{FPR}}, d_{\text{TYPR}})$ rule produced -2.821% relative gain for $p=50$. The conservative mean-distance rule improves this to +0.501% and beats all-D-optimal, which remains at -1.846%.

Strength: the measure turns sampling from a fixed rule into a data-driven transfer decision.

Limitation: Latin hypercube remains competitive in high-frequency/global-coverage cases; the current policy mixes only random and D-optimal points.

Next step: introduce a three-way policy over random, D-optimal, and Latin sampling or lower the D-optimal floor in far-distance cases.

7. Paper-Ready Interpretation

The experiment does not support the claim that measure-weighted sampling is universally superior to every design. It supports a more defensible claim: NN-PR/Taylor-PR distance is an actionable transfer gate for deciding when legacy PR/D-optimal technology can be used safely with a new NN model.